Lesson (1): Energy Transfer in Ecosystems

Get Ready!

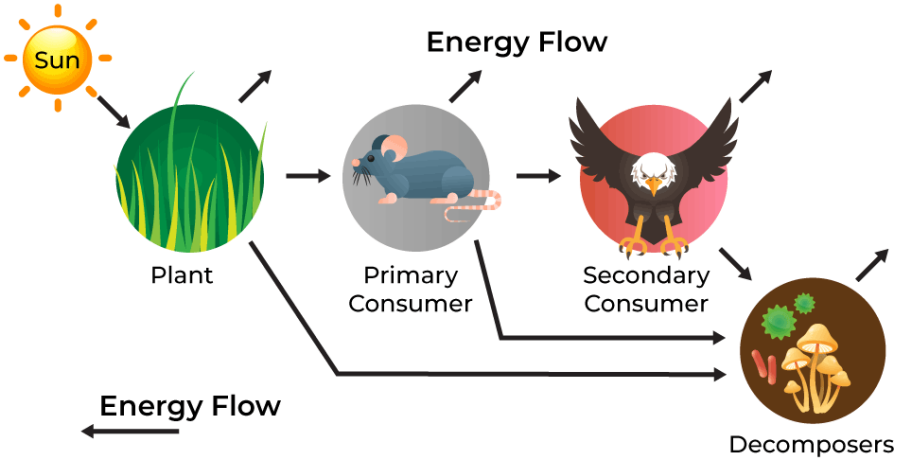
Imagine yourself in the heart of a dense forest, under the bright **sunlight**, where plants sparkle with vibrant **green hues** as they absorb **sunlight**. But how do these **plants** become a source of **food** for the **animals** living there?

What We will learn in this lesson?

- How energy from the sun flows through **ecosystems** via **food chains**
- How this process impacts **ecological balance**.
- Using concepts from **physics** and **chemistry**, we'll understand how **energy transfers**
- The effects of this transfer on **living organisms** in the **environment**.

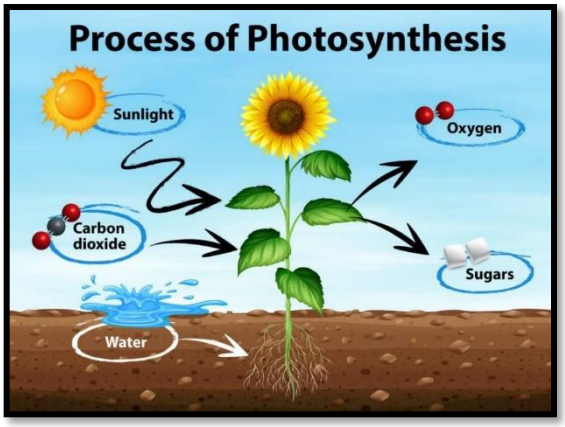
The Concept of Energy in an Ecosystem

One way to study the interaction between living organisms in ecosystems is by **tracing the transfer of energy within the system.**



Energy Transfer

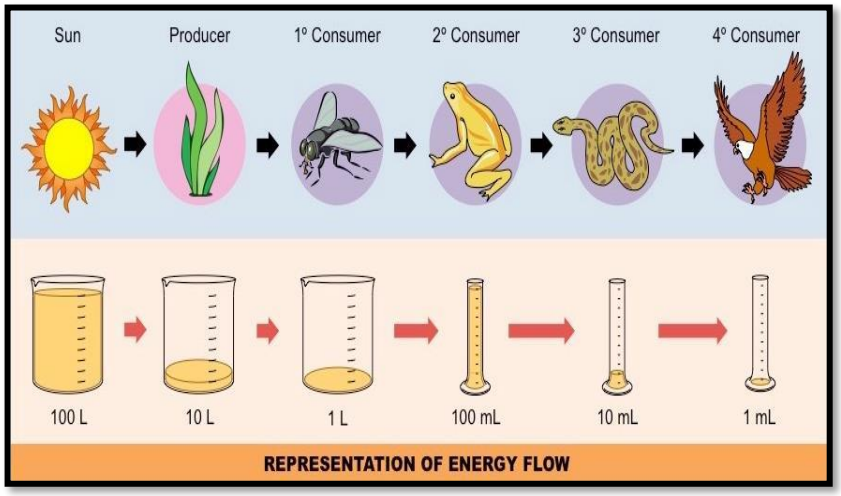
- When **sunlight** reaches plants, they perform **photosynthesis**.
- During this process, plants **convert solar energy into chemical energy** stored in sugar molecules within them.



$$\underset{\text{carbon dioxide}}{6CO_2} + \underset{\text{water}}{6H_2O} \xrightarrow{\text{Photosynthesis}} \underset{\text{glucose}}{C_6H_{12}O_6} + \underset{\text{oxygen}}{6O_2}$$

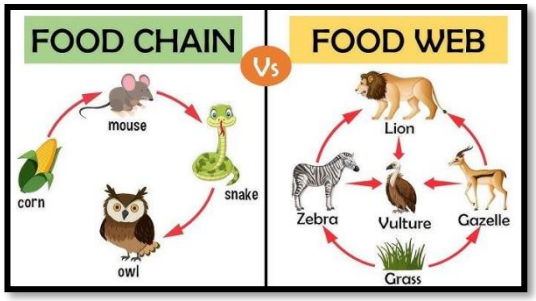
Energy Transfer Through Food Chains

- When insects feed on plants, they obtain the **chemical energy** stored in the plant's sugars.
- This energy is transferred from the plant to the **insect**
- When another animal feeds on the insect, the energy is transferred **from the insect to the animal**.
- With each transfer of energy between **trophic levels**, a significant amount of energy is **lost**.



The role of food chains and food webs

- **Ecologists** use food chains and food webs as models to illustrate energy transfer within an ecosystem.
  - Each step in the food chain or web is known as a **"trophic level."**



- Trophic levels begin with **autotrophic organisms**, which form the **first trophic level** in all ecosystems, followed by **heterotrophic organisms** that make up the **subsequent trophic levels.**

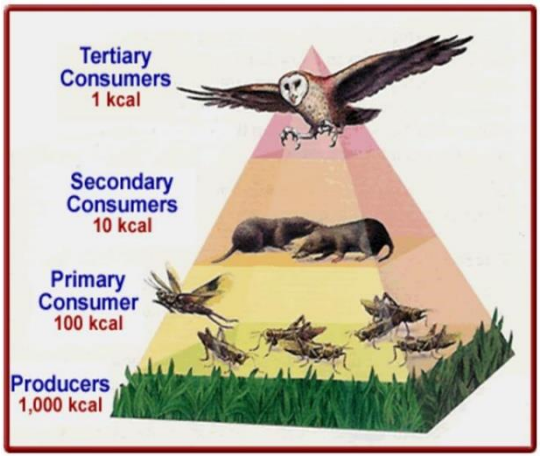
Measuring and transferring energy

• The **energy transfer pyramid** is used to illustrate the number of **organisms** relying on the same **energy source**, helping to determine the **efficiency of energy transfer** across trophic levels.

• The efficiency of energy transfer is approximately **10%** between each level (**explain**) about **90% of the energy** is **lost** in biological processes such as **excretion and respiration**, during which the largest amount of **glucose** (chemical energy) is consumed).

• Due to the significant energy loss, **energy pyramids** rarely contain more than **six levels**, as the remaining energy becomes insufficient to sustain another organism.

• Therefore, it is preferable for **humans** to consume **plants directly** rather than animals that feed on plants.



Calculating the lost energy

✓ **Lost energy** is the energy not directly used in biological processes.

✓ This does not contradict the **law of conservation of energy**; part of an organism's energy is lost as **heat** during **respiration** and other biological activities.

Illustrative Example: Suppose a **rabbit** obtains **100 joules (J)** of energy from consuming plants.

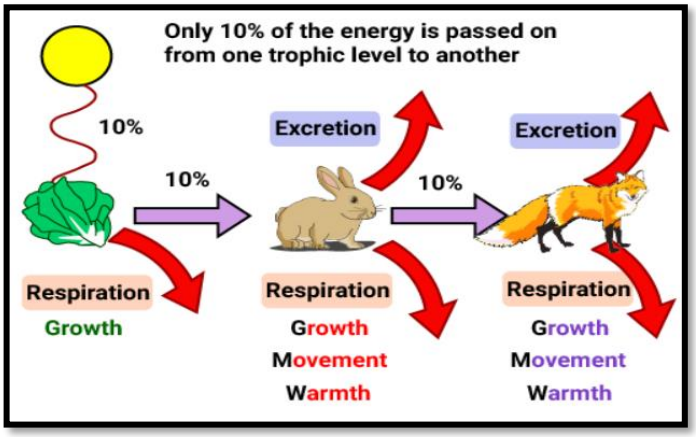
The energy is distributed as follows:

1. The majority is used during **cellular respiration**, where sugar is converted into **carbon dioxide (*chemical energy*)** and returned to the environment during exhalation.
2. Another portion is converted into **kinetic energy**, enabling the rabbit to **move**.
3. A portion is transformed into **thermal energy** to keep the body **warm**.
4. A small amount of energy remains in **undigested food (chemical energy)**, which is expelled as waste and returned to the soil through excretion.

✓ **When adding up these different forms of energy, we find that their total is 100 joules (J), which aligns with the law of conservation of energy.**

Lost energy

- ✓ It is simply the energy **not used directly**, representing the difference between the **total energy from food** and the **energy utilized by the organism** for essential processes such as movement and growth.



Scientific Activity

Suppose a plant receives **1000 joules (J)** of solar energy and uses **only 2%** of this energy in the process of **photosynthesis**, while the rest is lost as **heat, reflection, or absorption** in other parts.

Calculate:

1. The amount of energy the plant uses in **photosynthesis**.
2. The amount of energy that was **lost**.

Conservation of Energy

Law of conservation of Energy in Food Chains

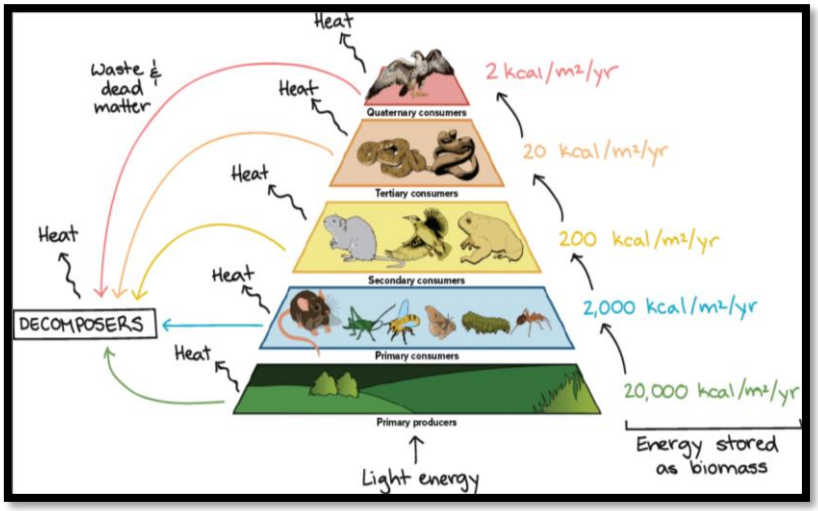
1. The **law of conservation of energy** is clearly demonstrated in **food chains** through the transformation of energy into various forms. The chain begins with **solar energy** from the sun, which is converted within **plants into chemical energy** stored in food during the process of **photosynthesis**.

2. This energy is then transferred to the **primary consumer** when it feeds on the plant. During this stage, the **chemical energy** undergoes conversion into **thermal** and **kinetic energy** through **respiration**, with some energy lost as heat.

3. When energy is transferred to the **secondary consumer**, which feeds on the **primary consumer**, additional energy loss occurs during processes such as **respiration** and excretion.

4. These **energy transformations** continue until the remaining chemical energy in the **decomposed organisms** is recycled into the soil by **decomposers** in the form of **nutrients**.

5. Throughout the food chain, energy is **lost** at each level in the form of **heat**. However, this does not contradict the **law of conservation of energy, which states that energy cannot be created or destroyed but only transformed from one form to another.**

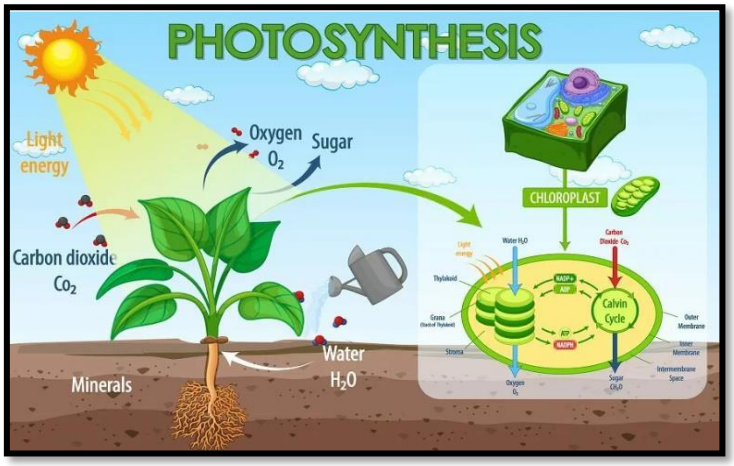


Chemistry in Energy Transfer

• The journey of energy transfer begins within living organisms, starting with **green plants.**

• During the process of photosynthesis, which takes place inside the **chloroplast (a specialized cellular organelle in plants).**

• **Complex chemical reactions take place.** The process begins with the absorption of **light by chlorophyll**, the green pigment in plants.



• Light stimulates chemical reactions that convert **carbon dioxide** and **water** into **glucose** and **oxygen**, as illustrated by the following equation:

$$\underset{\text{Carbon dioxide}}{6CO_2} + \underset{\text{Water}}{6H_2O} \xrightarrow[\text{Chlorophyll}]{\text{Light}} \underset{\text{Sugar}}{C_6H_{12}O_6} + \underset{\text{Oxygen}}{6O_2}$$

Photosynthesis ⇄ Respiration

• The **chemical energy** stored in **glucose** is utilized by other living organisms in the **food chain** when they feed directly or indirectly on plants.

• When **glucose** (a biofuel) is burned with **oxygen** inside the organism's body during the process of **respiration, thermal energy** is generated. This energy is essential for sustaining the life of the organism.

Respiration:

$$C_6H_{12}O_6 + 6O_2 \longrightarrow 6CO_2 + 6H_2O$$

• Thus, energy is transformed **from light energy into chemical energy** stored within the **chemical bonds** of the glucose molecule, and then into **thermal energy**, which contributes to the survival of living organisms.

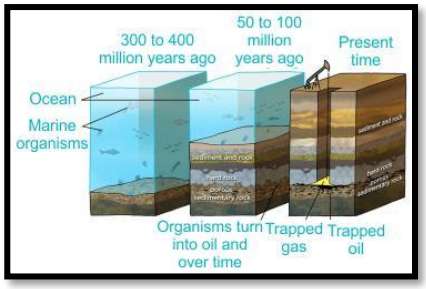
The energy stored within fossil fuels

Formation and Utilization of Fossil Fuels:

Formation of fossil fuels

- Fossil fuels such as **coal**, **petroleum**, and **natural gas** are formed from living organisms that stored **solar energy** either directly or indirectly.

Coal



- Composed primarily of **carbon (C)**.
- Formed from the remains of **decomposed trees and plants** buried deep within the Earth millions of years ago.

Petroleum

- A mixture of various **hydrocarbon compounds**.
- Originates from **marine organisms** and **plants** buried millions of years ago, decomposing under high **pressure** and **temperature**.

Natural Gas

- Consists of a mixture of **hydrocarbon gases**, primarily **methane (70–98%)**, with small amounts of **ethane**, **propane**, and **butane**.
- Found floating above **petroleum deposits** underground, within **coal mines**, or trapped between **rock layers**.

Utilization

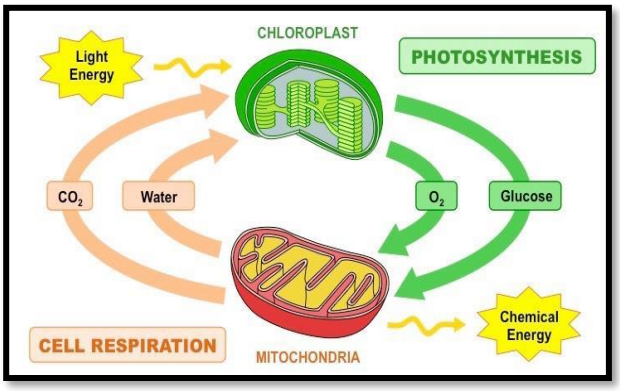
- When **fossil fuels** are burned with **oxygen**, such as in **internal combustion engines**, **thermal energy** is generated.
- This energy drives the **operation of machines** and is vital for their functionality.

The Relationship Between Photosynthesis and Cellular Respiration

• These **reactions** help maintain the balance of **oxygen** and **carbon dioxide** in the atmosphere.

• **Plants produce oxygen** and store energy in **glucose**.

• Meanwhile, other living organisms (such as **animals** and **humans**) **consume oxygen** and use **glucose** to produce energy, releasing **carbon dioxide** and **water** as waste products.



The Impact of the Relationship Between Photosynthesis and Cellular Respiration on the Ecosystem

1. Ecological Balance:

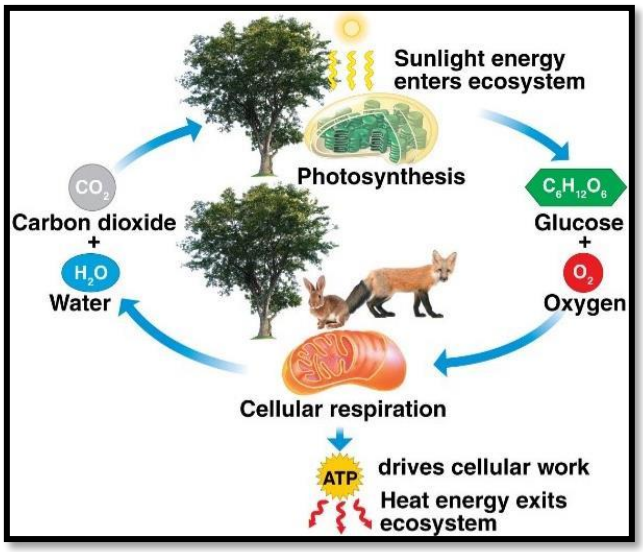
- The relationship between **photosynthesis** and **cellular respiration** contributes to maintaining the balance of **gases** in the atmosphere.
  - **Plants produce oxygen** and capture **carbon dioxide** during photosynthesis.
- Meanwhile, other living organisms **consume oxygen** and **produce carbon dioxide** during respiration, helping to maintain the **ecological balance**.

2. Energy Flow:

- The **solar energy** stored in **glucose molecules** through photosynthesis is transferred through the **food chain** when plants are consumed by animals.
- This energy is used in **cellular respiration** to produce **ATP (adenosine triphosphate)**, the primary energy source for all vital processes in living organisms.

3. The Carbon Cycle:

- **Photosynthesis** and **cellular respiration** both contribute to the natural **carbon cycle**, where carbon dioxide and water are **recycled** between the environment and living organisms.
- Simply put, photosynthesis provides **glucose** and **oxygen**, which are used in **cellular respiration** to release energy.
- This integration between photosynthesis and cellular respiration ensures the flow of **energy** and the balance of **gases** in the ecosystem, supporting life on Earth.



Research and Investigation

Practical Experiment: Exploring Photosynthesis

Materials Required:

1. An aquatic plant, such as **Elodea**
2. A **transparent bottle or glass cup** & water
3. **Baking soda (sodium bicarbonate)**
4. **Aluminum foil or opaque paper**
5. A **flashlight or light source** (e.g., sunlight)
6. **Paper and pens** for note-taking

Steps:

1. **Prepare the Plant:**
   - Fill the bottle or glass cup with **water**.
   - Add a small amount of **baking soda** (about half a teaspoon) to the water to increase the carbon dioxide level, which is essential for photosynthesis.
   - Place the **aquatic plant** in the water and allow it to adjust for a few minutes.
2. **Set Up the Light:**
   - If you're using a **flashlight**, direct the light toward the plant.
   - If you're using **sunlight**, place the plant in a sunny spot.
3. **Cover the Parts:**
   - To identify the areas where photosynthesis occurs, cover some of the plant's leaves with **aluminum foil or opaque paper**. Leave other leaves exposed to **sunlight** or **artificial light**.
4. **Observe the Reaction:**
   - Place the plant under the light and leave it for several hours.
   - After the period has passed, remove the aluminum foil or opaque paper. Observe any changes in the exposed leaves compared to the covered ones.
5. **Oxygen Test:**
   - Conduct a simple test for the presence of oxygen. Fill the glass cup with water, then place the plant in it. Wait for some time and observe the **air bubbles** that may form on the surface of the water. These bubbles are the result of **oxygen production** during photosynthesis.
6. **Analysis and Interpretation:**
   - Compare the results between the exposed and covered leaves. Observe where the most bubbles were produced—exposed leaves should produce more oxygen than the covered ones, indicating that **photosynthesis** occurs in the exposed leaves.
   - Write your observations about how **light affects the photosynthesis process**.